

# Component-based system reliability subject to positive and negative correlation

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## ABSTRACT

Traditional system reliability modeling methods assume statistical independence of component failures, while past research emphasizes methods to characterize positive correlation and dependence. However, negative dependence and correlation are also important for system reliability and resilience modeling. This paper develops a method to assess the reliability of a system subject to arbitrary combinations of positive and negative correlation between the failures of its components. The method enables derivation of expressions composed of a vector of unequal component reliabilities and unequal pairwise component correlations. The method is illustrated through a sequence of examples, which conduct sensitivity analysis on correlation parameters of the model as well as quantify the impact of component correlations on system reliability. Coverage experiments formally assess the flexibility and scalability of the approach.

## 1. Introduction

Classical reliability methods assume component failures exhibit statistical independence [1], whereas research considering dependence and correlation [2] are often limited to systems with few components and positive dependence or correlation. Such assumptions ease model formulation and solution complexity. However, systems can experience failures that are either negative or positively correlated [3]. Depending on the structure, correlated failures can lower or increase system reliability. Moreover, with the growing importance of resilience modeling, there is practical interest in methods to quantify the impact of negative correlation to guide the design of systems that will be robust to disruptions. For example, geospatially distributing major access points of the internet backbone to improve fault tolerance to natural disasters may create negative correlations that lower the probability users experience poor quality of service. Therefore, scalable methods are needed for systems susceptible to combinations of positive and negative correlation.

Several approaches to model positive dependence and correlation have been proposed previously. This literature review is restricted to common mathematical methods employed to model correlation and dependence and is not intended to provide an exhaustive summary of the many extensions and modeling applications to which these methods have been applied. Lehmann [4] was the first to define positive dependence. Dozens of system modeling papers employ the Markov

method [5] to characterize failure dependence behaviors. Chae and Clark [6] proposed the concept of common-cause failures among distinct subsets of components with life distributions. The notion of positive quadrant dependence was introduced by Joe [7], while Vaurio's "implicit method" [8] models failure correlation in subsets of components. More recent CCF models include the work of Zhou et al. [9] who proposed a CCF model that utilizes condition monitoring data on centrifugal pumps in order to superimpose the impact of maintenance on the evolution of degrading components. Kim et al. [10] proposed an approach to model CCFs for multi-unit probabilistic safety assessment involving a large number of nuclear power plant units, while Shekhar et al. [11] presented a Markovian warm-spare model of nodes in a fault tolerant computing network to provision resources when spare nodes are subject to switching failure and common cause failure.

Additional common methods include copulas, shock processes, and correlated distributions. For example, Limbourg [12] employed copulas to encode spatial dependence and its impact on the reliability of a fault tolerant system, while Wang and Pham [13] used copulas in a s-dependent competing risk model experiencing multiple degradation processes as well as fatal and nonfatal random shocks. Zhang et al. [14] proposed a copula-based stress-strength time-varying correlation interference model for structural reliability analysis. Matus et al. [15] proposed an optimization procedure to match the marginal failure probabilities and failure correlations of the Marshall-Olkin copula for telecommunication network design and reliability analysis despite an

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| Notation              |                                              |                   |                                                                      |
|-----------------------|----------------------------------------------|-------------------|----------------------------------------------------------------------|
| $n$                   | Number of components                         | $\rho_{i,j}$      | Correlation between failures of component $i$ and $j$                |
| $\mathbf{R}_n$        | Vector of variables denoting component state | $\rho_{i,j}^-$    | Lower bound on correlation between failures of component $i$ and $j$ |
| $R_i$                 | State of component $i$                       | $\rho_{i,j}^+$    | Upper bound on correlation between failures of component $i$ and $j$ |
| $E[\mathbf{R}]$       | Reliability of component vector              | $IP_{\rho_{i,j}}$ | Improvement potential correlation $\rho_{i,j}$                       |
| $\mu_i$               | Reliability of component $i$                 | $B_{\phi_i}$      | Birnbaum measure of parameter $i$                                    |
| $\Sigma_{n \times n}$ | Component failure covariance matrix          |                   |                                                                      |

exponential number of parameters. Fang et al. [16] presented a class of bivariate stochastic processes incorporating environmental stress variables and model the dependence between two degradation processes with a copula function, demonstrating the models use on light-emitting diodes and polymeric materials. Ling et al. [17] employed Gumbel-Hougaard and Frank copulas to analyze one-shot devices, such as munitions, rockets, and automobile airbags, possessing two correlated failure modes with data collected from constant-stress accelerated life tests. Safaei et al. [18] studied optimal age replacement policies for parallel and series systems with dependent components characterized by copulas. Li et al. [19] proposed a system reliability analysis method based on a hybrid of multivariate Gaussian process and univariate Gaussian process models to respectively model groups of highly interdependent and relatively independent components observed in cantilever beam-bar mixed systems and vehicle side crashes.

Anastasiadis et al. [20] modeled aging and statistical dependence where damage accumulation is multiplicative with each shock. Reliability models and preventive maintenance policies were developed by Peng et al. [21] for systems experiencing multiple dependent competing failures and shock processes. Arnold et al. [22] considered correlated failures by treating component failures as shocks which damage other components. Meango and Ouali [23] developed a model for dependent failure modes as consequences of extreme shocks and demonstrated its application to a culvert system subject to two cracking and displacement dependent failure modes. Zhao and Xing [24] proposed a combinatorial hierarchical methodology for reliability analysis of a smart home sensor system subject to cascading probabilistic functional dependence and random failure propagation time. Fiondella and Zee-phongsekul [25] developed a modeling approach for  $k$ -out-of- $n$  systems constructed from correlated but identical components, while Fiondella and Zee-phongsekul [26] derived a trivariate Bernoulli distribution with unequal success probabilities and unequal correlations to assess the impact of positive correlation on fault tolerant software.

Studies have also developed methods to explicitly quantify the impact of negative dependence or correlation. Joag-Dev and Proschan [27] first discussed the concept of negative association. The monograph of Lai and Xie [2] offers a systematic survey of dependence, including negative dependence. However, it is acknowledged in [2] that few models easily scale to systems composed of more than three components. Fiondella and Zee-phongsekul [28] also developed a model of recovery block fault tolerant software as a function of the versions and acceptance test reliabilities and the correlations between these components. A numerical method was proposed by Fiondella [29] for unequal component reliabilities and unequal positive and negative component correlation. However, the approach does not scale because it requires solution of  $2^n$   $n$ -variable integrals. Moreover, algebraic generalization was not possible because this approach was based on the multivariate normal distribution, which lacks a closed-form solution. Moreover, while many copula-based techniques can also consider negative correlation, Liu et al. [30], constructed rotated Clayton copulas to explicitly model positive and negative correlation, demonstrating improved performance over several alternative copulas ability to track crude oil futures price and prices of a Chinese stock index futures.

This paper presents computationally efficient methods to derive analytical reliability expressions with explicit correlation parameters as

well as exact numerical methods for reliability assessment of systems possessing unequal component reliabilities and unequal positive and negative component correlations. We assess the scalability of the approach with a sequence of illustrations, demonstrating its potential for efficient reliability and sensitivity analysis. We also examine the limitations and tradeoffs between the algebraic and numerical approaches as well as their flexibility.

Section 2 summarizes the conditional linear family to simulate a sequence of correlated Bernoulli random variables. Section 2.1 develops the analytical generalization, while Section 3 presents an efficient numerical approach. Section 4 discusses sensitivity analysis and importance assessment, while Section 5 provides illustrations. Section 6 concludes with suggestions for additional research.

## 2. Modeling

This section describes the conditional linear family (CLF) [28], which is subsequently generalized to the analytical case and applied to several component based systems.

Let  $\mathbf{R}_{n \times 1}$  denote the states of a vector of components with  $\mathbf{R}_n = \langle R_1, \dots, R_n \rangle^T$ .  $R_i = 1$  if component  $i$  is reliable and  $R_i = 0$  otherwise. For simplicity, success and failure are alternatively denoted by  $R_i$  and  $\bar{R}_i$  with  $E[\mathbf{R}] = \langle \mu_1, \dots, \mu_n \rangle^T$ , which may be interpreted as the probability a component in a discrete system is reliable or the probability that a component survives for a given duration such as a warranty period in a continuous system. The correlation between a pair of components  $i$  and  $j$  is denoted  $\rho_{i,j}$ . The covariance matrix  $\Sigma$  possesses entries  $\Sigma_{i,j} = \rho_{i,j} \sigma_i \sigma_j$ , where  $\sigma_i = \sqrt{\text{Var}[R_i]} = \sqrt{\mu_i \bar{\mu}_i}$ ,  $\bar{\mu}_i = 1 - \mu_i$ . For  $i = 2, \dots, n$ , let  $\mathbf{R}_{i-1} = \langle R_1, \dots, R_{i-1} \rangle^T$  be the observed success and failure of the previous  $(i-1)$  components. Given  $\mathbf{R}_{i-1}$ , the conditional linear family [31,32], expresses success of component  $i$  as

$$E[R_i | \mathbf{R}_{i-1}] = \mu_i + \sum_{j=1}^{i-1} (R_j - \mu_j) b_j, \quad (1)$$

where  $\mathbf{b}_{(i-1) \times 1} = \Sigma_{i-1}^{-1} \boldsymbol{\tau}_i$ ,  $\Sigma_{i-1} = \text{Cov}(\mathbf{R}_{i-1}, \mathbf{R}_{i-1})$  is the upper left square matrix of dimension  $(i-1) \times (i-1)$  and  $\boldsymbol{\tau}_i = \text{Cov}(\mathbf{R}_{i-1}, R_i)$  is the  $(i-1) \times 1$  vector to the right of  $\Sigma_{i-1}$  characterizing the covariance between the failures of the  $i$ th component and the preceding  $(i-1)$  components,  $(2 \leq i \leq n)$ . Furthermore,  $E[\bar{R}_i | \mathbf{R}_{i-1}] = 1 - E[R_i | \mathbf{R}_{i-1}]$ .

Fig. 1 shows possible paths the CLF approach can take with two components  $R_1$  and  $R_2$ . Commencing from the Start node, component one ( $R_1$ ) is reliable or unreliable with probability  $\mu_1$  or  $1 - \mu_1$  respectively. However, component one influences the second component ( $R_2$ ) through correlation  $\rho_{1,2}$ .

Fig. 1 indicates conditional dependence with the labels on the edges between realizations of  $R_1$  and  $R_2$ . For example,  $E[R_2 | R_1]$  indicates the probability  $R_2$  is reliable given that  $R_1$  was reliable. Computing conditional probabilities from Eq. (1) requires  $\mu_1$ ,  $\rho_{1,2}$  and the information that component one was reliable. This deconditioning step produces outcomes of the joint distribution.

If  $E[R_i | \mathbf{R}_{i-1}] \notin (0, 1)$  on any path, then the procedure fails to encode an otherwise valid combination of component reliabilities and correlations. With the exception of sampling directly from the complete joint distribution, there are few or no methods capable of simulating all valid combinations of component reliabilities and correlations. Therefore, in

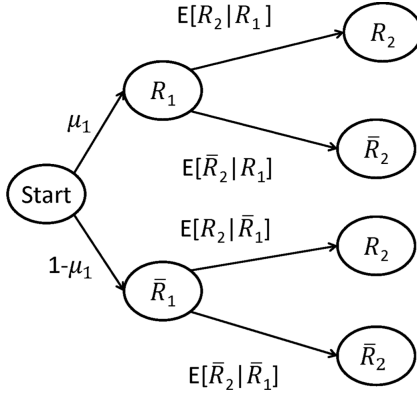


Fig. 1. Realization tree of correlated components.

addition to reliability and sensitivity analysis as well as performance, the illustrations also consider the flexibility of the proposed method with respect to the parameter space, measured according to the fraction of the valid parameter space covered by the method, including both positive and negative correlation.

### 2.1. Algebraic generalization

In this section, the simulation algorithm discussed above is generalized to the algebraic case. This process produces expressions for the reliability of two component series and parallel systems including an explicit correlation parameter. We subsequently generalize to three and then  $n$  component systems.

#### 2.1.1. Two components

Expressions for two components are derived by computing the success probability of component two given success in the first component. Since  $\Sigma_1^{-1} = \langle 1/\sigma_1^2 \rangle$  and  $\tau_1 = \langle \rho_{1,2}\sigma_1\sigma_2 \rangle$ ,  $\mathbf{b}_1 = \langle \rho_{1,2}\sigma_2/\sigma_1 \rangle$  and

$$E[R_2|R_1 = 1] = \mu_2 + \frac{\bar{\mu}_1\rho_{1,2}\sigma_2}{\sigma_1}, \quad (2)$$

It follows from conditional probability that

$$E[R_2|R_1] = \frac{E[R_1R_2]}{E[R_1]}, \quad (3)$$

Hence, multiplying Eq. (2) by  $E[R_1] = \mu_1$  yields

$$E[R_1R_2] = \mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2, \quad (4)$$

It follows from the algebra of expectations that

$$E[R_1\bar{R}_2] = \mu_1 - E[R_1R_2], \quad (5)$$

$$E[\bar{R}_1R_2] = \mu_2 - E[R_1R_2], \quad (6)$$

and

$$E[\bar{R}_1\bar{R}_2] = 1 - \mu_1 - \mu_2 + E[R_1R_2]. \quad (7)$$

Setting  $\rho_{1,2} = 0$  in Equations 4–(7) reduces to the case [33], where components one and two are independent. It is possible to solve the Pearson product-moment correlation coefficient equation to obtain  $E[R_1R_2]$  [34] and therefore Eqs. (4)–(7). However, deconditioning is also applicable to more than two variables, enabling expressions for systems possessing three or more components.

Two component reliability expressions

It follows From Eq. (4) that the reliability of the two component series is

$$E[R_s] = \mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2, \quad (8)$$

while the two component parallel system reliability expression can be obtained from the complement of Eq. (7)

$$E[R_p] = \mu_1 + \mu_2 - (\mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2). \quad (9)$$

#### 2.1.2. Three components

Similar to the two component case, we find

$$\mathbf{b}_i = \frac{\sigma_3 \left( \rho_{i,3} - \frac{\prod_{j < k} \rho_{j,k}}{\rho_{i,3}} \right)}{\sigma_i(1 - \rho_{1,2}^2)}, \quad i \in \{1, 2\}, \quad (10)$$

so that  $\Pr\{R_3 = 1|R_1 = r_1, R_2 = r_2\}$  is

$$\mu_3 + \frac{\sigma_3}{1 - \rho_{1,2}^2} \sum_{i=1}^2 (-1)^{\bar{r}_i} \left( \rho_{i,3} - \frac{\prod_{j < k} \rho_{j,k}}{\rho_{i,3}} \right) \times \sqrt{\frac{\mu_i^{\bar{r}_i} \bar{\mu}_i^{r_i}}{\mu_i^{r_i} \bar{\mu}_i^{\bar{r}_i}}}, \quad (11)$$

where  $\bar{r}_i = 1 - r_i$  and  $\Pr\{R_3 = 0|R_1 = r_1, R_2 = r_2\}$  is

$$1 - \Pr\{R_3 = 1|R_1 = r_1, R_2 = r_2\}$$

To the authors' knowledge [35], an explicit parameterization has not been proposed in the literature.

Three component reliability expressions

Expressions for the reliability of three component systems may be obtained from Eq. (11) as follows.

The reliability of a three component series is

$$\begin{aligned} E[R_s] &= E[R_1R_2R_3] = (\mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2)(\mu_3 + \bar{\mu}_1b_1 + \bar{\mu}_2b_2) \\ &= \mu_3 + \frac{\sigma_3}{1 - \rho_{1,2}^2} \sum_{i=1}^2 \left( \rho_{i,3} - \frac{\prod_{j < k} \rho_{j,k}}{\rho_{i,3}} \right) \times \sqrt{\frac{\bar{\mu}_i}{\mu_i}}, \end{aligned} \quad (12)$$

where  $E[R_1R_2R_3] = \Pr\{R_1 = 1, R_2 = 1, R_3 = 1\}$ .

The reliability of a 2-out-of-3 good system is

$$\begin{aligned} E[R_{2,3}] &= E[\bar{R}_1R_2R_3] + E[R_1\bar{R}_2R_3] + E[R_1R_2\bar{R}_3] + E[R_1R_2R_3] \\ &= (\bar{\mu}_1\mu_2 - \rho_{1,2}\sigma_1\sigma_2)(\mu_3 - \bar{\mu}_2b_{1,1} + \mu_1b_{2,1}) \\ &\quad + (\mu_1\bar{\mu}_2 - \rho_{1,2}\sigma_1\sigma_2)(\mu_3 - \mu_2b_{1,1} + \bar{\mu}_1b_{2,1}) + (\mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2), \end{aligned} \quad (13)$$

where  $E[R_1R_2\bar{R}_3] + E[R_1\bar{R}_2R_3]$  simplifies to  $E[R_1R_2] = (\mu_1\mu_2 + \rho_{1,2}\sigma_1\sigma_2)$ .

Similarly, the reliability of a parallel system composed of three components is

$$\begin{aligned} E[R_p] &= 1 - E[\bar{R}_1\bar{R}_2\bar{R}_3] \\ &= 1 - (\bar{\mu}_1\bar{\mu}_2 + \rho_{1,2}\sigma_1\sigma_2)(\bar{\mu}_3 + \mu_1b_{1,1} + \mu_2b_{2,1}) \\ &= 1 - \left[ \mu_3 - \frac{\sigma_3}{1 - \rho_{1,2}^2} \sum_{i=1}^2 \left( \rho_{i,3} - \frac{\prod_{j < k} \rho_{j,k}}{\rho_{i,3}} \right) \times \sqrt{\frac{\mu_i}{\bar{\mu}_i}} \right]. \end{aligned} \quad (14)$$

#### 2.1.3. N-Component reliability expressions

The probability of a given vector of outcomes  $\mathbf{R}_n$  is

$$E \left[ \prod_{i=1}^n R_i \right] = \prod_{i=1}^n \left( \mu_i + \sum_{j=1}^{i-1} b_{ij} \bar{\mu}_j^{\bar{R}_j} \mu_j^{R_j} \right). \quad (15)$$

A proof is given in Appendix A.

Eq. (15) can also be written in matrix form because the  $(i, j)$ th entry of  $\Sigma_{n-1}^{-1}$  in  $\mathbf{b} = \Sigma_{n-1}^{-1}\tau_1$  may be expressed according to the minors of the matrix or  $\frac{-\det(\Sigma_{n-1})_{i,j}}{\det(\Sigma_{n-1})}$ , where  $\det(\Sigma_{n-1})_{i,j}$  is the determinant of  $\Sigma_{n-1}$  with row  $i$  and column  $j$  removed. The diagonal elements are  $\frac{\det(\Sigma_{n-1})_{i,i}}{\det(\Sigma_{n-1})}$ .

### 3. Numerical method

The conditional linear family provides the necessary framework to assess the reliability of structured systems, but can be optimized for efficiency by noting that not all outcomes of the joint distribution need to be enumerated. For example, referring to Fig. 1, the two component series system is only reliable when both components are reliable. Therefore, knowledge of the cut sets of the system enables an optimized numerical algorithm that only enumerates outcomes which contribute

to system reliability, avoiding the calculation associated with the failure of the first component ( $\bar{R}_1$ ) and subsequent nodes as well as failure of component two given reliable operation of component one ( $E[\bar{R}_2|R_1]$ ). Similar logic can be applied to the two component parallel system and any system structure representable by a reliability block diagram or fault tree. Moreover, path sets can enable early termination at a node in the realization tree where all subsequent outcomes are no longer relevant. For example, the two component parallel system can stop after calculating the success probability of the first component because this is sufficient for the parallel system to be reliable, but will also require calculation of success in component two when component one fails ( $E[R_2|\bar{R}_1]$ ).

**Algorithm 1** summarizes an efficient recursive numerical method *NumericalReliability* to compute the reliability of an  $n$  component system, which enumerates only outcomes relevant to the system based on the cut sets  $C$  and optimizes based on the path sets  $\mathcal{P}$ , both of which are determined from the system structure function  $\phi$ .

The initial invocation of the recursive procedure *NumericalReliability* ( $T, Rel$ ) begins at the root of the realization tree  $T = \{\}$  corresponding to the empty path prior to consideration of success or failure in the first component with system reliability initialized to  $Rel \leftarrow 0$ . Since the system under consideration is a binary state system, the branching factor at each node in the realization tree is two and we denote the outcome of a reliable component as the left child ( $T \rightarrow Left$ ) and the outcome of an unreliable component with the right child ( $T \rightarrow Right$ ). The two conditions on line one correspond to instances where the algorithm has reached a node that contributes to system reliability. In the first case, the length of the path after success ( $T \rightarrow Left$ ) is  $n$ . Since the path was not pruned previously because of a prior cut set, the system must be reliable, assuming it is a coherent system. In the second case, the path sets are compared to the sequence of component success and failures through  $T \rightarrow Left$ . If either of these conditions is satisfied, the node contributes to system reliability and probability of the path added to  $Rel$ . If neither of these conditions is satisfied, then success and failure from the next component must be enumerated. Line four expands the successful outcome of the next component, while line six first tests if failure in the present component contains a cut. If not, line seven expands the outcomes of the next component recursively.

#### 4. Sensitivity analysis and importance assessment

This section explains the valid range of correlation between a pair of component reliabilities. It also develops the reliability improvement potential and Birnbaum measures [36] to quantify the impact of reducing correlations within a system. Generally speaking, engineering techniques to reduce the negative impact of correlation on a system will be domain specific. Moreover, there may be more than one approach to mitigate the impact of correlation. Furthermore, each technique may incur a corresponding cost. Therefore, sensitivity analysis and importance assessment techniques can be used to assess the relative effectiveness of two or more alternative strategies to mitigate the negative impact of correlation on a system.

##### 4.1. Range of correlation

Prentice [37] showed that  $\rho_{j,k}$  cannot always take values in the interval  $[-1, 1]$ . Restrictions are necessary to ensure all outcomes of the joint component reliability distribution reside in the interval  $[-1, 1]$ . Specifically, the lower and upper bounds of correlation  $\rho_{j,k}$  are functions of the reliabilities of components  $j$  and  $k$

$$\rho_{j,k}^- = \max\left(-\sqrt{\frac{\mu_j\mu_k}{\bar{\mu}_j\bar{\mu}_k}}, -\sqrt{\frac{\bar{\mu}_j\bar{\mu}_k}{\mu_j\mu_k}}\right) \quad (16)$$

$$\rho_{j,k}^+ = \min\left(\sqrt{\frac{\mu_j\mu_k}{\bar{\mu}_j\bar{\mu}_k}}, \sqrt{\frac{\bar{\mu}_j\bar{\mu}_k}{\mu_j\mu_k}}\right). \quad (17)$$

##### 4.2. Improvement potential

Traditionally, the improvement potential assesses the utility of achieving a perfectly reliable component  $i$ . Because the models produced by the proposed approach include explicit correlation parameters it is also possible to apply this measure to assess the benefit of eliminating the correlation between a pair of components such that  $\rho_{i,j} = 0$ . Thus, the improvement potential of a correlation is

$$IP_{\rho_{i,j}} = R(E[\mathbf{R}], \Sigma|\rho_{i,j} = 0) - R(E[\mathbf{R}], \Sigma), \quad (18)$$

The right hand term is an estimate of system reliability given the component reliabilities  $E[\mathbf{R}]$  and covariance matrix  $\Sigma$ , while the left hand term in Eq. (18) sets correlation  $\rho_{i,j}$  to zero. Hence, the improvement potential of a correlation parameter quantifies the reliability increase that could be attained by eliminating correlation between components  $i$  and  $j$ .

While eliminating correlation entirely (or reducing to  $\rho_{i,j}^-$ ) may not be feasible in all engineered systems, this measure quantifies the utility of an alternative design in which the correlation does not exist. This measure will also be useful in instances where concepts from reliability theory are employed to quantify system resilience, where removing the correlation between two components represents a system specific enhancement such as geospatial diversity or forms of component or subsystem shielding.

##### 4.3. Birnbaum measure

Analytical reliability models can also identify component improvements that would improve system reliability most significantly. The Birnbaum measure [38],

$$B_{\theta_i} = \frac{\partial R(\Theta)}{\partial \theta_i} \Big|_{\Theta=\hat{\Theta}}, \quad (19)$$

is commonly employed to assess alternative system reliability optimization strategies and is calculated by computing the partial derivative with respect to parameter  $\theta_i \in \Theta$  and evaluating this with numerical values  $\hat{\Theta}$  for the present parameter estimates. The Birnbaum measure quantifies the improvement that could be attained from an infinitesimal increase in parameter  $\theta_i$ . Sorting the Birnbaum measure of each parameter in descending order can be used to identify component reliability improvements that should be prioritized in order to cost effectively allocate limited resources that will optimize system reliability despite constraints.

Correlation parameters of a reliability or resilience model can also

**Require:** Cut sets  $C(\phi)$

**Require:** Path sets  $\mathcal{P}(\phi)$

**Require:** Number of components  $n$

**Require:** Component reliabilities  $E[\mathbf{R}]$

**Require:** Component correlations  $\Sigma$

```

1: if PathLength( $T \rightarrow Left$ ) ==  $n$  or Path( $T \rightarrow Left$ ) then
2:    $Rel \leftarrow Rel + PathProbability(T \rightarrow Left)$ 
3: else
4:    $NumericalReliability(T \rightarrow Left, Rel)$ 
5: end if
6: if not Cut( $T \rightarrow Right$ ) then
7:    $NumericalReliability(T \rightarrow Right, Rel)$ 
8: end if
```

**Algorithm 1.** *NumericalReliability*( $T, Rel$ ) .



be included in assessments with the Birnbaum measure. However, we seek to assess the system reliability improvement obtainable by decreasing correlation. The Birnbaum measure is therefore negated to quantify the improvement in system reliability that could be attained by decreasing correlation.

## 5. Illustrations

This section illustrates the utility of the algebraic, numerical, and simulation methods. Section 5.1 presents applications to reliability and sensitivity analysis, while Section 5.2 compares the performance of these alternative approaches. Section 5.3 conducts coverage analysis to assess the flexibility of the approach to assess the reliability of systems with different component reliabilities and correlations.

### 5.1. Reliability and sensitivity analysis

For the sake of exposition, we conduct sensitivity analysis in the context of two and three component systems and reliability analysis on a five component bridge network.

#### 5.1.1. Two component series system

Fig. 2 shows correlation's influence on series system reliability specified in Eq. (8) for four combinations of component reliabilities  $\mu_1$  and  $\mu_2$ , where the correlation bounds have been determined from Eqs. (16) and (17) respectively. Fig. 2 indicates that increasing correlation increases series system reliability, which agrees with intuition, since an increase in the correlation between two components increases the probability both components will be reliable, thus increasing series system reliability. Fig. 2 also suggests that system reliability is equivalent to the less reliable component when correlation is maximized ( $\rho_{1,2}^+$ ), which also agrees with intuition since the component possessing lower reliability always operates when the more reliable component is operational. At the other extreme ( $\rho_{1,2}^-$ ), negative correlation in a series system reliability equals  $\min(1 - (\bar{\mu}_1 + \bar{\mu}_2), 0)$ , since an extreme negative correlation maximizes outcomes where one component is reliable but the other fails, including  $E[\bar{R}_1 \bar{R}_2]$  and  $E[R_1 \bar{R}_2]$ . This reduces overlap between the failure of both components ( $E[\bar{R}_1 \bar{R}_2]$ ) as well as success in both ( $E[R_1 R_2]$ ) components of the series system.

To assess the trends in Figs. 2 and 3 shows correlation's influence on the four outcomes of two components when  $\mu_1 = 0.4$  and  $\mu_2 = 0.9$ . Fig. 3 shows that an increase in correlation increases the outcomes  $E[R_1 R_2]$  and  $E[\bar{R}_1 \bar{R}_2]$ . Hence, an increase in correlation also increases the probability both components work or fail. Since  $E[R_1 R_2]$  is the single outcome which contributes to the reliability of a series system, an increase in correlation also increases the reliability of a series. However, an increase in correlation also decreases the outcomes, where one component fails but the other component operates ( $E[\bar{R}_1 R_2]$ ,  $E[R_1 \bar{R}_2]$ ).

Fig. 4 shows the influence of correlation and component reliability on series system reliability, where the reliability of component one is a value greater or less than 0.5. Specifically,  $\mu_1 = \{0.8, 0.3\}$  are considered for the sake of illustration. The reliability of component two is varied in the range  $0 \leq \mu_2 \leq 1$  and the system reliability is calculated over the valid range of correlation defined by Eqs. (16) and (17). Both surfaces in Fig. 4 show that an increase in component two's reliability or the correlation increases series system reliability, since increasing component reliability increases coherent system reliability and an increase in correlation also increases the probability that both components are reliable, as observed in Fig. 3, thereby increasing system reliability.

#### 5.1.2. Two component parallel system

Fig. 5 shows correlation's influence on the reliability of a two component parallel system defined in Eq. (9).

Like Fig. 2, the limits on correlation vary according to Eqs. (16) and (17). Unlike Fig. 2, however, Fig. 5 shows that an increase in

correlation decreases parallel system reliability. Intuitively, increasing correlation should increase the probability both components fail. Furthermore, Fig. 3 indicates that outcome  $E[\bar{R}_1 \bar{R}_2]$  increases, contributing to unreliability in the parallel system.

Fig. 5 also shows that the reliability of a parallel system equals the reliability of the more reliable component when correlation is maximized ( $\rho_{1,2}^+$ ), since the component possessing lower reliability only operates when the more reliable component operates, totally defeating the purpose of fault tolerance. Conversely, the parallel system reliability at the minimum correlation ( $\rho_{1,2}^-$ ) is  $\max(\mu_1 + \mu_2, 1)$ , since overlap between the cases where both components are reliable is minimized.

Negative correlation improves the reliability of a parallel system because this negative correlation decreases the probability both components fail simultaneously. Furthermore, when the sum of the two component reliabilities is greater than one, maximizing negative correlation leads to a system reliability of 1.0, such that one component is operational when the other fails. While it may not be possible to attain this theoretical limit in real-world engineered system, the approach presented here offers a method to quantify attainable improvements in reliability or resilience through negative correlation.

Fig. 6 shows the influence of correlation and component reliability on parallel system reliability when the first component's reliability is held constant at  $\mu_1 = \{0.8, 0.3\}$ . Fig. 6 shows that an increase in component reliability leads to an increase in the reliability of a parallel system, while an increase in correlation leads to a decrease in the reliability of a parallel system.

#### 5.1.3. Three component systems

Figs. 7–9 shows the impact of correlation on three component series, parallel, and 2-out-of-3 systems, given in Eqs. (12), (14), and (13) respectively for parameters  $E[\mathbf{R}] = \langle 0.8, 0.75, 0.7 \rangle$  and  $\rho_{1,2} = 0.1378$ ,  $\rho_{1,3} = 0.0326$ ,  $\rho_{2,3} = 0.1109$ . Figs. 7–9 vary  $\rho_{12}$ ,  $\rho_{13}$ , and  $\rho_{23}$  within their feasible range according to Eqs. (16) and (17). These figures indicate that increasing correlation decreases the reliability of the parallel and 2-out-of-3 systems but increases the reliability of the series system.

##### Improvement potential

Table 1 summarizes the improvement potential defined in Eq. (18) for the series, 2-out-of-3, and parallel system. Table 1 indicates that setting  $\rho_{1,2}$  to zero, decreases the series system reliability by 1.5%, but increases the reliability of the 2-out-of-3 and parallel system. These trends correspond with Fig. 7 and provide an exact quantification of the change in reliability when  $\rho_{1,2}$  decreases from 0.1378 to 0.0. The latter rows of Table 1 provide a similar assessment for the correlations between components one and three and components two and three respectively.

##### Birnbaum Measure

Table 2 summarizes the Birnbaum measure defined in Eq. (19) for each of the three systems with respect to each correlation parameter. Table 2 indicates that a very small reduction in correlation would lower series reliability, but increase the 2-out-of-3 and parallel system

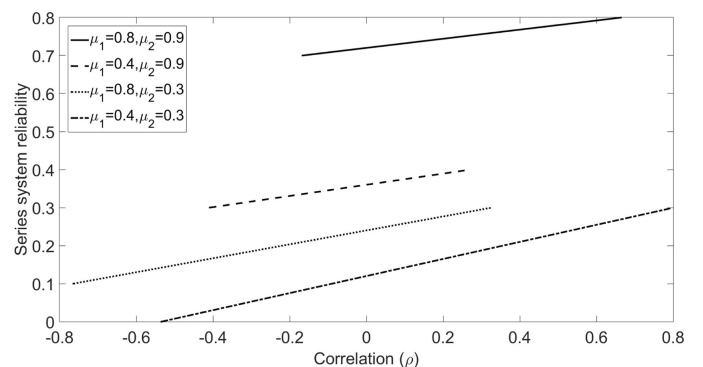


Fig. 2. Influence of correlation on series system.

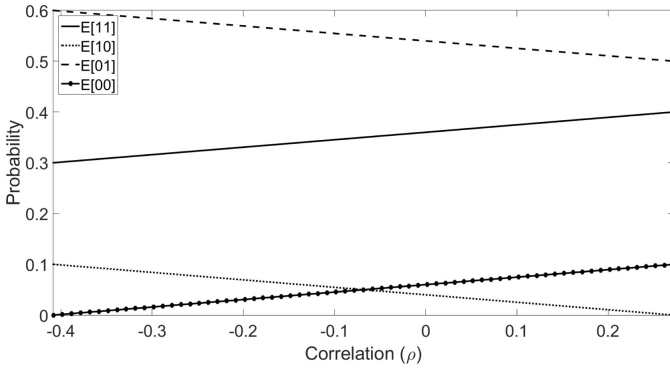


Fig. 3. Influence of correlation on outcomes of joint distribution.

reliability. Moreover, the decrease in  $\rho_{2,3}$  would be most detrimental to the series because components two and three are the least reliable, whereas the decrease in  $\rho_{1,2}$  benefits the parallel system most because components one and two are the most reliable.

#### 5.1.4. Bridge network

To illustrate the application of Eq. (15), Fig. 10 shows a bridge network composed of five components.

The complete joint distribution possesses  $2^5 = 32$  outcomes. However, only combinations of component outcomes that provide a path from the start to destination node are considered reliable.

Fig. 11 shows the realization tree of the bridge. Combinations of success or failure in a subset of components that lead to system reliability or unreliability are denoted with a ✓ or x respectively. For example, the failure of components one and two is a cut set of the bridge and the outcomes of components three, four, and five need not be considered further. Similarly, success in components one through four contain the path set consisting of components one and four, rendering the outcome of component five irrelevant.

The reliability of the bridge can be expressed as the summation of the eleven leaf nodes in Fig. 11 possessing a ✓ such that

$$\begin{aligned}
 E[R_b] = & E[R_1 R_2 R_3 R_4] + E[R_1 R_2 R_3 \bar{R}_4 R_5] + E[R_1 R_2 \bar{R}_3 R_4] \\
 & + E[R_1 R_2 \bar{R}_3 \bar{R}_4 R_5] + E[R_1 \bar{R}_2 R_3 R_4] + E[R_1 \bar{R}_2 R_3 \bar{R}_4 R_5] \\
 & + E[R_1 \bar{R}_2 \bar{R}_3 R_4] + E[\bar{R}_1 R_2 R_3 R_4] + E[\bar{R}_1 R_2 R_3 \bar{R}_4 R_5] \\
 & + E[\bar{R}_1 R_2 \bar{R}_3 R_4 R_5] + E[\bar{R}_1 \bar{R}_2 R_3 \bar{R}_4 R_5]
 \end{aligned} \quad (20)$$

For the sake of illustration,

$$E[\mathbf{R}] = \langle 0.8, 0.75, 0.70, 0.72, 0.73 \rangle$$

with pairwise component failure correlations

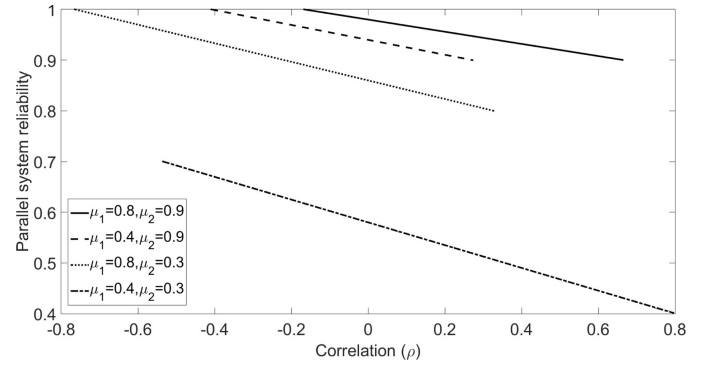


Fig. 5. Influence of correlation on parallel system.

|     |          |          |          |           |
|-----|----------|----------|----------|-----------|
| 1.0 | 0.137777 | 0.032623 | 0.071992 | -0.048682 |
|     | 1.0      | 0.110859 | 0.098119 | 0.180018  |
|     |          | 1.0      | 0.087384 | 0.130338  |
|     |          |          | 1.0      | 0.024884  |
|     |          |          |          | 1.0       |

Table 3 reports the numerical probabilities of the eleven outcomes contributing to bridge reliability identified in Fig. 11.

Thus, the reliability of the bridge considering these unequal component reliability and unequal pairwise positive and negative correlations is  $E[R_b] = 0.838043$ . The corresponding value of the bridge when correlations are set to zero is 0.857168. To understand why the bridge with no correlations is more reliable, note that the correlations between the parallel paths in components one and two as well as the correlation between components four and five are positive, which decreases the probability that one path is available when the other is not. The approach thus offers a method to precisely quantify the impact of arbitrary combinations of unequal component reliabilities, unequal positive and negative correlations, and system structure on reliability.

#### 5.2. Performance analysis

This section performs a comparative analysis of the algebraic and numerical approaches on systems composed of anywhere from two to eight components. For the sake of illustration,

$$E[\mathbf{R}] = \langle 0.8, 0.75, 0.70, 0.72, 0.73, 0.77, 0.67, 0.5 \rangle$$

with pairwise component failure correlations used in the bridge example as well as the following additional correlations for components six through eight

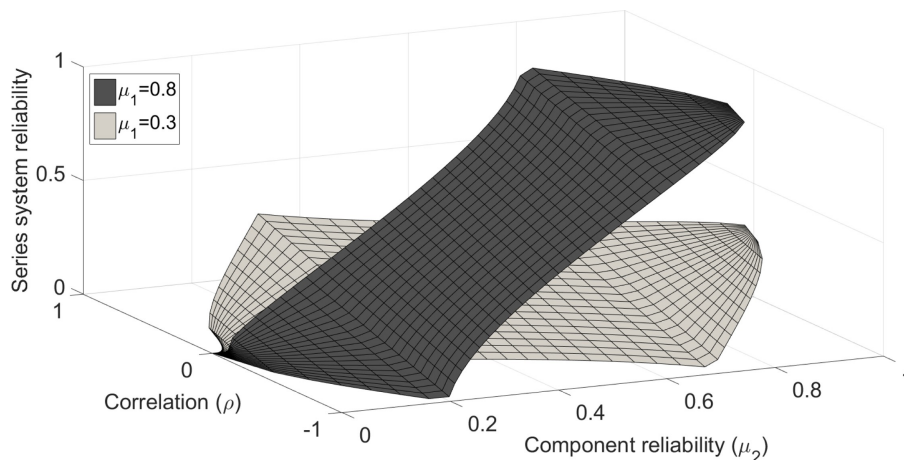


Fig. 4. Influence of correlation and component reliability on series system.

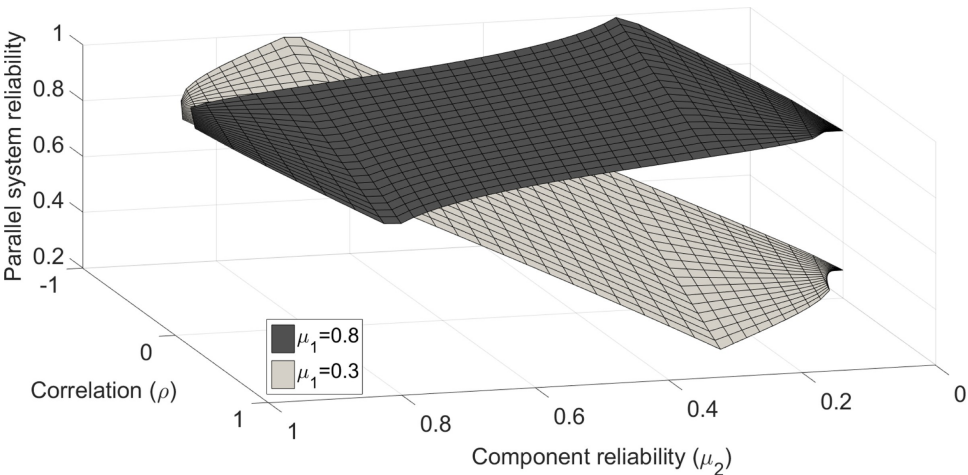


Fig. 6. Influence of correlation and component reliability on parallel system.

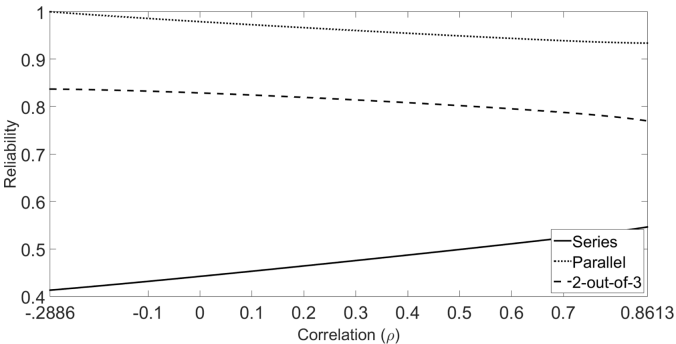


Fig. 7. Sensitivity to  $\rho_{1,2}$ .

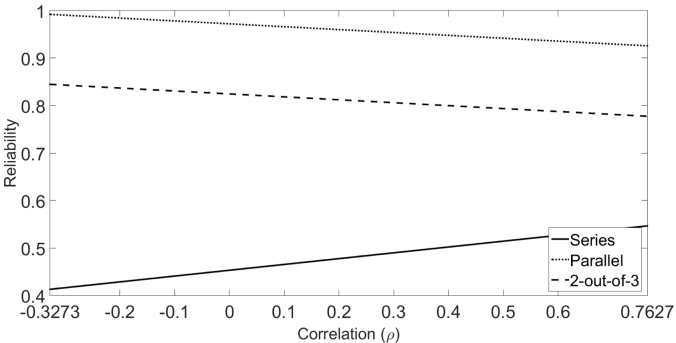


Fig. 8. Sensitivity to  $\rho_{1,3}$ .

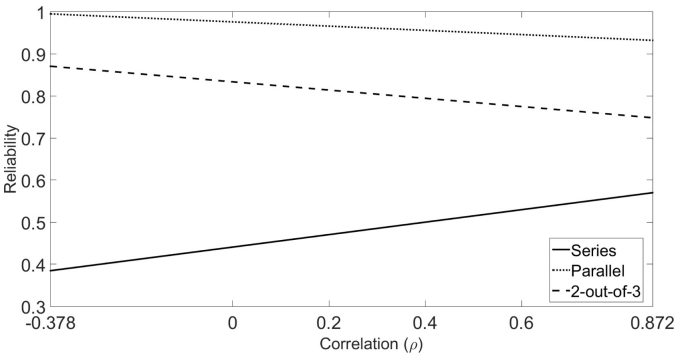


Fig. 9. Sensitivity to  $\rho_{2,3}$ .

Table 1  
Improvement potential.

| Correlation  | Series   | 2-out-of-3 | Parallel |
|--------------|----------|------------|----------|
| $\rho_{1,2}$ | – 0.0150 | 0.0062     | 0.0088   |
| $\rho_{1,3}$ | – 0.0039 | 0.0020     | 0.0019   |
| $\rho_{2,3}$ | – 0.0164 | 0.0108     | 0.0056   |

Table 2  
Birnbaum measure.

| Correlation  | Series   | 2-out-of-3 | Parallel |
|--------------|----------|------------|----------|
| $\rho_{1,2}$ | – 0.1110 | 0.0488     | 0.0622   |
| $\rho_{1,3}$ | – 0.1225 | 0.0617     | 0.0608   |
| $\rho_{2,3}$ | – 0.1482 | 0.0979     | 0.0502   |

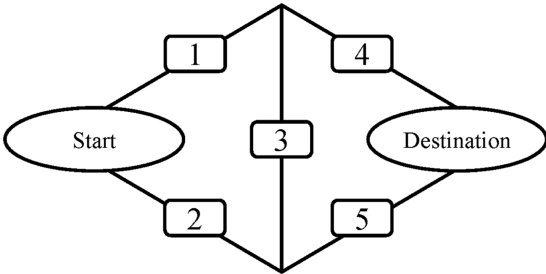


Fig. 10. Bridge structure.

$$\begin{pmatrix} 0.091320 & -0.066384 & 0.017863 \\ 0.016341 & -0.059739 & -0.034677 \\ 0.116186 & -0.056192 & -0.040215 \\ -0.032124 & -0.000254 & 0.028627 \\ -0.006967 & 0.127017 & -0.016643 \\ & 1.0 & 0.014421 & -0.014069 \\ & & 1.0 & -0.030581 \\ & & & 1.0 \end{pmatrix}$$

Table 4 reports the reliability as well as the time (in seconds) taken to derive algebraic expressions for  $n$ -component series systems, the time required to evaluate these expressions with the first  $n$  component reliabilities and correlations to obtain numerical estimates, and the time taken to perform numerical evaluation according to Algorithm 1. Table 4 divides the time to perform algebraic calculations into the symbolic derivation and numerical evaluation steps, which respectively correspond to the calculations required to obtain the expression and time to substitute numerical values into the expression respectively.

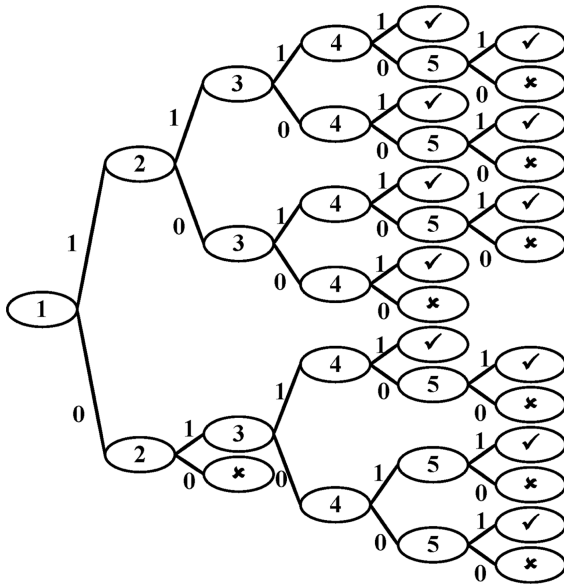


Fig. 11. Realization tree of bridge.

**Table 3**  
Probability of outcomes contributing to bridge reliability.

| Outcome (R)                          | Pr{R}    | Outcome (R)                                | Pr{R}    |
|--------------------------------------|----------|--------------------------------------------|----------|
| $E[R_1 R_2 R_3 R_4]$                 | 0.355053 | $E[\bar{R}_1 \bar{R}_2 \bar{R}_3 R_4]$     | 0.041527 |
| $E[R_1 R_2 R_3 \bar{R}_4 R_5]$       | 0.080542 | $E[\bar{R}_1 R_2 R_3 R_4]$                 | 0.063925 |
| $E[R_1 R_2 \bar{R}_3 R_4]$           | 0.117017 | $E[\bar{R}_1 R_2 R_3 \bar{R}_4 R_5]$       | 0.022691 |
| $E[R_1 R_2 \bar{R}_3 \bar{R}_4 R_5]$ | 0.033793 | $E[\bar{R}_1 R_2 \bar{R}_3 R_4 R_5]$       | 0.017734 |
| $E[R_1 \bar{R}_2 R_3 R_4]$           | 0.075333 | $E[\bar{R}_1 R_2 \bar{R}_3 \bar{R}_4 R_5]$ | 0.010091 |
| $E[R_1 \bar{R}_2 R_3 \bar{R}_4 R_5]$ | 0.020337 |                                            |          |

**Table 4**  
Comparative performance of algebraic and numerical approaches on series system (in seconds).

| Components | Reliability | Algebraic derivation | Algebraic evaluation | Numerical evaluation |
|------------|-------------|----------------------|----------------------|----------------------|
| 2          | 0.623864    | 0.000                | 0.000                | 0.000                |
| 3          | 0.457129    | 0.001                | 0.000                | 0.001                |
| 4          | 0.355053    | 0.123                | 0.001                | 0.003                |
| 5          | 0.281250    | 3.157                | 0.006                | 0.028                |
| 6          | 0.226299    | 3.420                | 0.055                | 0.028                |
| 7          | 0.151700    | 7.827                | 0.484                | 0.028                |
| 8          | 0.072392    | 452.304              | 5.243                | 0.003                |

**Table 5**  
Comparative performance of algebraic and numerical approaches on parallel system (in seconds).

| Components | Reliability | Algebraic derivation | Algebraic evaluation | Numerical evaluation |
|------------|-------------|----------------------|----------------------|----------------------|
| 2          | 0.926136    | 0.000                | 0.000                | 0.000                |
| 3          | 0.970287    | 0.001                | 0.000                | 0.001                |
| 4          | 0.986682    | 0.127                | 0.001                | 0.001                |
| 5          | 0.994446    | 2.319                | 0.006                | 0.025                |
| 6          | 0.997684    | 3.007                | 0.007                | 0.027                |
| 7          | 0.999396    | 7.676                | 0.473                | 0.027                |
| 8          | 0.999663    | 518.907              | 5.198                | 0.003                |

This division was performed to illustrate the utility of the algebraic method because the derivation only needs to be performed once, whereas the evaluation step will need to be performed for each set of unique parameter values, thus enabling the efficient sensitivity analysis

shown in the previous section on two and three component systems.

Although slower, algebraic derivation makes some unique forms of sensitivity analysis possible such as differentiation to calculate the Birnbaum measure (Eq. (19)) with respect to both component reliability and correlation parameters. Moreover, the numerical evaluation of the four component series system took 1/123rd the time to obtain the reliability expression and the time to evaluate the five component series was 526 (3.157/0.006) times faster than the derivation.

Table 4 indicates that algebraic derivation requires only a few seconds for seven or fewer components. This suggests that algebraic derivation is most suitable for computing the Birnbaum measure of systems composed of seven or fewer components, but Algorithm 1 is more appropriate for calculating the improvement potential and reliability of systems.

Table 5 reports similar results for  $n$ -component parallel systems.

Performance of algebraic derivation as well as algebraic and numerical evaluation are similar to the results for series systems because parallel system reliability is calculated as one minus the probability of all components failing, which imposes similar computational requirements to calculation of all components succeeding in series.

To assess the efficiency of the algebraic and numerical approaches on systems possessing multiple outcomes that lead to a reliable system, Table 6 reports results for  $n$ -component  $k$ -out-of- $n$  and bridge systems.

### 5.3. Coverage analysis

To objectively assess the flexibility of the proposed approach, three coverage tests were performed on the systems considered in the previous section. The first of these tests, referred to henceforth as  $T_1$  was a uniform test of coverage of the parameter space. In this test, each component reliability was selected uniformly at random from the range  $\mu_i \in (0.001, 0.999)$ , then pairwise correlations were generated uniformly at random from the interval defined by Eqs. (16) and (17). The efficient numerical procedure was then executed and considered successful if  $E[R_i | \mathbf{R}_{i-1}] \in (0, 1)$  at all intermediate steps. This test was performed 10,000 times for each system and the number of fraction of successful runs recorded.

The second and third tests are related and hence referred to as Test  $T_{2A}$  and Test  $T_{2B}$ . These tests were originally described in [26], which utilized data from Knight and Leveson [39] to assess the coverage of an analytical expression for a 2-out-of-3 system composed of highly reliable but positively correlated versions. Hence, our previous study [26] only applied the tests to an analytical expression for 2-out-of-3 systems with correlations restricted to the range  $(0, \rho_{ij}^+)$  because of limitations in the underlying encoding technique of the method reported there.

The formulation and execution of tests  $T_{2A}$  and  $T_{2B}$  are explained in detail here. The reliability distribution of the versions (components) reported in Ref. [39] are characterized by a beta distribution with parameters  $\alpha = 1,000$  and  $\beta = 2$ , according to the theoretical framework of Littlewood and Miller [40], in order to approximate the average and spread of version reliabilities. Moreover, the empirical CDF of the 162 correlations between pairs of versions (components) was constructed. In test  $T_{2A}$ , component reliabilities are sampled from the beta distribution and correlations from the empirical CDF. If a sampled correlation violated the bounds  $(\rho_{ij}^-, \rho_{ij}^+)$  the experiment was immediately considered as failing the coverage test, whereas  $T_{2B}$  resampled from the empirical CDF until the bounds were satisfied. The efficient numerical method presented in Section 3 was then executed and consider a success if  $E[R_i | \mathbf{R}_{i-1}] \in (0, 1)$  on all paths because the procedure was able to encode a valid combination of component reliabilities and correlations. Both tests  $T_{2A}$  and  $T_{2B}$  were executed 100,000 times and the fraction of successful runs interpreted as the coverage of the method.

Table 7 reports the results of the coverage experiments on series systems composed of between two and eight components. Table 7 indicates that, for all three tests, coverage decreases as the number of components



**Table 6**

Comparative performance of algebraic and numerical approaches on  $k$ -out-of- $n$  and bridge system (in seconds).

| System     | Reliability | Algebraic derivation | Algebraic evaluation | Numerical evaluation |
|------------|-------------|----------------------|----------------------|----------------------|
| 2-out-of-3 | 0.822584    | 0.003                | 0.001                | 0.001                |
| 3-out-of-5 | 0.847982    | 2.156                | 0.093                | 0.002                |
| Bridge     | 0.838043    | 2.251                | 0.095                | 0.002                |

**Table 7**

Comparative analysis of coverage tests on series system.

| Components | Test $T_1$ | Test $T_{2A}$ | Test $T_{2B}$ |
|------------|------------|---------------|---------------|
| 2          | 1.0000     | 0.9912        | 1.0000        |
| 3          | 0.8498     | 0.9736        | 0.9993        |
| 4          | 0.6266     | 0.9482        | 0.9974        |
| 5          | 0.3945     | 0.9136        | 0.9938        |
| 6          | 0.2076     | 0.8738        | 0.9875        |
| 7          | 0.0967     | 0.8261        | 0.9789        |
| 8          | 0.0414     | 0.7752        | 0.9680        |

**Table 8**

Comparative analysis of coverage tests on parallel system.

| Components | Test $T_1$ | Test $T_{2A}$ | Test $T_{2B}$ |
|------------|------------|---------------|---------------|
| 2          | 1.0000     | 0.9917        | 1.0000        |
| 3          | 0.8455     | 0.9737        | 0.9982        |
| 4          | 0.6162     | 0.9445        | 0.9936        |
| 5          | 0.3874     | 0.9053        | 0.9852        |
| 6          | 0.2096     | 0.8616        | 0.9727        |
| 7          | 0.1018     | 0.8090        | 0.9571        |
| 8          | 0.0440     | 0.7489        | 0.9370        |

**Table 9**

Comparative analysis of coverage tests on  $k$ -out-of- $n$  and bridge systems.

| Components | Test $T_1$ | Test $T_{2A}$ | Test $T_{2B}$ |
|------------|------------|---------------|---------------|
| 2-out-of-3 | 0.7548     | 0.9462        | 0.9706        |
| 3-out-of-5 | 0.0226     | 0.7022        | 0.7634        |
| Bridge     | 0.0296     | 0.7196        | 0.7796        |

increases. Coverage decreases most rapidly in test  $T_1$ . However, this corresponds to the most general case where the component reliabilities range in the interval  $\mu_i \in (0.001, 0.999)$ . Since it is unlikely that a system would be constructed from components with uniformly random reliabilities and correlations, this result may be regarded as the more theoretical statistical coverage of the numerical method for a product of correlated random variables. Test  $T_{2A}$  and  $T_{2B}$  exhibit better coverage. While the coverage is only 77.52% for an eight component series in test  $T_{2A}$ , we recall that this results is for the most general case of unequal component reliabilities and arbitrary combinations of unequal positive and negative correlations. Moreover, past methods [2] do not scale well

beyond more than three components and the only guaranteed method to attain 100% coverage is to utilize the entire joint distribution, but that the availability of such information may prevent this alternative in many scenarios. Thus, even the conservative assessment of coverage embodied in test  $T_{2A}$  extends the applicability to systems composed of up to eight components and the less stringent test  $T_{2B}$  exhibits coverage greater than 95%.

Table 8 reports coverage experiments on parallel systems composed of between two and eight components. The results are both quantitatively and qualitatively similar to those observed for the series system.

Table 9 reports the results of the coverage experiments on  $k$ -out-of- $n$  and bridge systems. Both tests  $T_{2A}$  and  $T_{2B}$  exhibit excellent coverage on the 2-out-of-3 system and fair coverage on the 3-out-of-5 and bridge system. Thus, the proposed approach offers meaningful improvements in the scalability and flexibility of methods to assess systems composed of correlated components.

## 6. Conclusions and future research

This paper presented analytical and numerical method to quantify the impact of correlated component failures on the reliability of a system possessing arbitrary structure, unequal component reliabilities, and unequal positive and negative component failure correlations. The algebraic generalization was developed for two, three, and  $n$  components systems. The examples illustrated reliability and sensitivity analysis in the context of two component series and parallel systems three component series, 2-out-of-3, and parallel systems as well as reliability analysis of the five component bridge. Performance and coverage analysis respectively assessed the scalability and flexibility of the approach on larger series and parallel system as well as the  $k$ -out-of- $n$  and bridge systems.

Future research will attempt to develop alternative methods in order to further enhance scalability and flexibility. Potential methods include: (1) reordering variables to consider those that occur most frequently in a system's path or cutsets first, (2) approximate methods that terminate search early, but maintain an error bound, and (3) importance sampling to enable efficient simulation techniques.

## CRedit authorship contribution statement

**Bentolhoda Jafary:** Investigation, Formal analysis, Visualization, Software, Writing - original draft. **Andrew Mele:** Investigation, Visualization, Software, Validation, Writing - original draft. **Lance Fiondella:** Conceptualization, Supervision, Methodology, Writing - original draft, Writing - review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Proof of N-component reliability expressions

Eq. (15) is proved by induction by first noting that substituting  $n = 1$  produces  $\mu_1$ . Furthermore,

$$\begin{aligned} E[\prod_{i=1}^{n+1} R_i] &= \prod_{i=1}^n \left( \mu_i + \sum_{j=1}^{i-1} b_j \bar{\mu}_j^{R_j} \mu_j^{\bar{R}_j} \right) \times \left( \mu_{n+1} + \sum_{j=1}^n b_j \bar{\mu}_j^{R_j} \mu_j^{\bar{R}_j} \right) \\ &= \prod_{i=1}^{n+1} \left( \mu_i + \sum_{j=1}^{i-1} b_j \bar{\mu}_j^{R_j} \mu_j^{\bar{R}_j} \right), \end{aligned} \quad (A.1)$$

where  $\left( \mu_{n+1} + \sum_{j=1}^n b_j \bar{\mu}_j^{R_j} \mu_j^{\bar{R}_j} \right)$  is the inductive hypothesis.

## Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.res.2020.107058](https://doi.org/10.1016/j.res.2020.107058)

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